FAKE JOB POSTING DETECTION

Visualization, Explainable AI & Hybrid Machine Learning for Recruitment Fraud Prevention

A Comparative Study & Unified System Design based on Recent Research in Online Recruitment Fraud Detection

Seminar / Project Presentation

Abstract

The Problem

- Online recruitment platforms have simplified hiring, but have also enabled a sharp rise in fraudulent job postings that cause financial loss, identity theft and emotional distress.
- This work consolidates four complementary research directions into one system view: (1) data visualization & text clustering of fake postings, (2) explainable multilingual transformer-based detection, (3) a hybrid ML + rule-based + API verification system (JobShield), and (4) an ANN + TF-IDF classification and recommendation engine.

The Convergent Insight

- Together these studies show that fraudulent postings share detectable signatures — vague language, unrealistic salary/ benefit promises, missing interviews, generic customer-facing roles and geographic clustering.
- Combining EDA/clustering, transformer-based multilingual classification, rule-based red-flag checks, external verification and Explainable AI (SHAP/LIME) yields a system that is both highly accurate and transparent to end users.

Project Motivation

Why automated fake job detection matters now

US $450M

lost by Americans to fake job scams (FTC, 2024)

AU $2.74B

total financial loss reported in Australia

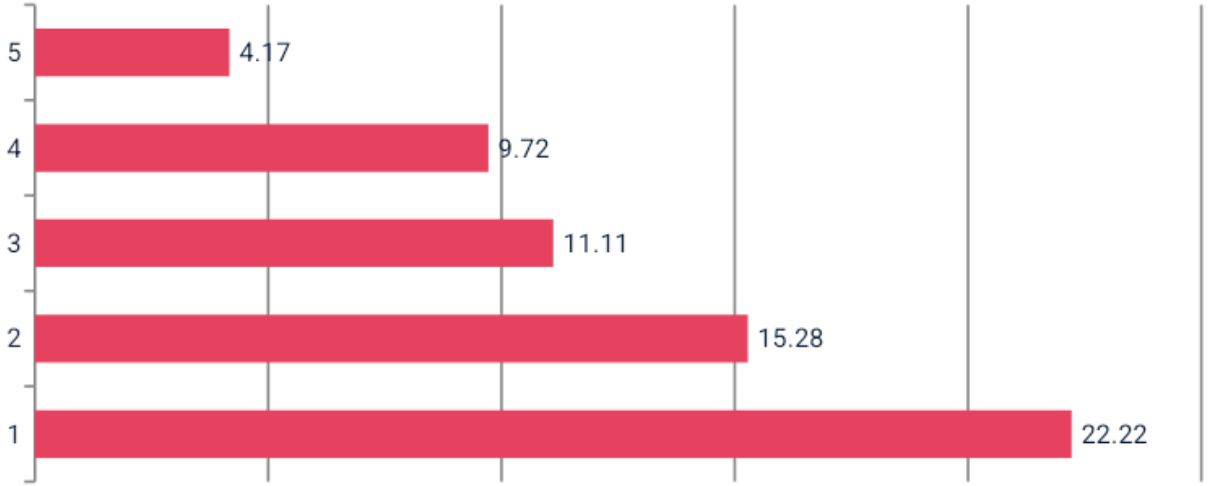
150%

rise in Australian job-scam losses vs. 2022

12%

of sampled Workable postings were fraudulent (EMSCAD)

Fraudulent postings by industry (EMSCAD, % of fake job postings)



Why this matters

- Manual review cannot scale to thousands of daily postings across global platforms
- Scams disproportionately target vulnerable groups: students, entry-level and flexible-work seekers
- Fraudulent postings mimic legitimate ones closely, defeating simple keyword filters
- An accurate, explainable, and multilingual detection system builds trust in online recruitment

Literature Survey (I)

Machine learning & deep learning approaches

Study / Model	Technique	Reported Accuracy
Naive Bayes / SVM / Logistic Regression	Classical ML on textual features (TF-IDF)	80 – 89%
Random Forest / AdaBoost / Gradient Boosting	Ensemble classifiers, GridSearchCV tuning	Above 98% (reported)
Sequential NN + GloVe, Bi-LSTM, GRU/Attention-GRU	Deep learning on job description sequences	Above 97% (reported)
Joseph & Mathew — ANN classifier	Multi-layer ANN on TF-IDF features	High accuracy (paper-reported)
Zhang — Bi-LSTM	Contextual sequence modelling	High accuracy, high complexity
MBERT / XLM-RoBERTa (Explainable AI study)	Multilingual transformer classification	97.8% (XLM-RoBERTa, best)
JobShield — Random Forest (hybrid)	TF-IDF + Random Forest + rule checks	87.4% (F1 = 0.86)

Literature Survey (II)

Explainability, hybrid systems & text visualization

1 Explainable Multilingual Detection

mBERT & XLM-RoBERTa classify postings in English, Tamil, Hindi, Telugu, Bengali; SHAP & LIME expose the words driving each prediction (e.g. “registration fee”, “urgent hiring”).

2 JobShield — Hybrid Fraud Detection

Combines a Random Forest classifier (TF-IDF features) with rule-based red-flag checks and external API verification (company, phone, domain) — accuracy 85–90%.

3 Visualization & Clustering (EMSCAD)

Tableau EDA + Orange hierarchical clustering reveal that fraud concentrates in customer-facing roles, Oil & Energy / Real Estate industries, and specific U.S. cities (Austin, Dallas, Bakersfield).

4 ANN + Recommendation Engine

TF-IDF features feed an Artificial Neural Network; beyond a real/fake label the system recommends apply/avoid actions and a fraud risk level (Low/Medium/High).

Existing Systems & Limitations

Existing Systems

- Classical ML (Naive Bayes, SVM, Logistic Regression) on TF-IDF text features
- Ensemble models (Random Forest, AdaBoost, Gradient Boosting) for higher accuracy
- Deep learning: Bi-LSTM, GRU, attention-based sequence models
- Single-language transformer / classification pipelines deployed as web apps (e.g. Streamlit)
- EDA & clustering used mainly for descriptive insight, not real-time prediction

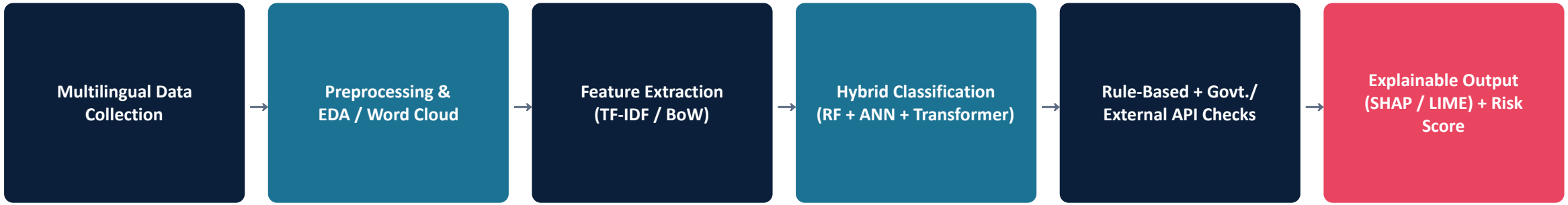
Key Limitations

- Function largely as “black boxes” — no explanation for why a posting is flagged
- Rely on text analysis only; no verification of recruiter identity or company legitimacy
- Most models trained on a single, imbalanced, English-only dataset (EMSCAD)
- Struggle with multilingual and code-mixed job postings (regional languages)
- High-accuracy deep/transformer models demand heavy compute and frequent retraining

Proposed System — Overview

A unified, explainable, hybrid fraud-detection pipeline

The proposed system merges the strengths of all four studies into a single pipeline: descriptive EDA/clustering for insight, multilingual transformer + ANN/ Random-Forest classification for accuracy, rule-based red-flag checks and external API verification for precision, and SHAP/LIME explainability for user trust.



Design goals

- High accuracy across languages and formats
- Transparent, human-readable explanations
- Real-time government registration check + external data verification
- Actionable recommendations (apply / avoid, risk level)
- Scalable to large, growing recruitment platforms
- Continuous learning as new scam patterns emerge

Proposed System — Key Innovations

1

Multilingual Coverage

English, Hindi, Tamil, Telugu, Bengali & code-mixed text via mBERT / XLM-RoBERTa

2

Hybrid Decision Logic

ML/DL prediction + rule-based red flags (Gmail-only recruiter, WhatsApp-only contact, unrealistic salary)

3

External + Government Verification

Checks if the company is government-registered (MCA / GST / Companies House / SEC EDGAR) plus phone & domain checks via Twilio, WhoisXML, OpenCage

4

Explainable Predictions

SHAP (global) & LIME (local) highlight the exact words driving each Real/Fake decision

5

Text Clustering & EDA

Hierarchical clustering groups postings by theme, exposing scam sub-types (e.g. “Client Relations Scam”)

6

Actionable Recommendations

Risk score (Low/Medium/High) plus apply/avoid guidance and verification tips for users

Software & System Requirements

Software Requirements

- Language: Python 3.9+
- ML / DL: scikit-learn, TensorFlow / Keras, PyTorch, HuggingFace Transformers (mBERT, XLM-RoBERTa)
- NLP: NLTK / spaCy for tokenization, lemmatization, stop-word filtering
- Explainability: SHAP, LIME
- Visualization / EDA: Tableau, Orange Data Mining, Matplotlib / Seaborn
- Deployment: Streamlit / Flask web app
- External & Govt. APIs: OpenCage, Twilio, OpenCorporates, WhoisXML, MCA21 / GST (India), Companies House (UK), SEC EDGAR (US)
- Database: MySQL / MongoDB for postings & prediction logs

Hardware / System Requirements

- Processor: Intel i5 / AMD Ryzen 5 or higher (i7 recommended for transformer training)
- RAM: 8 GB minimum, 16 GB+ recommended for transformer fine-tuning
- GPU: CUDA-enabled GPU (e.g. NVIDIA RTX series) recommended for mBERT / XLM-RoBERTa training
- Storage: 20 GB+ free space for datasets, embeddings and model checkpoints
- OS: Windows 10/11, Linux (Ubuntu) or macOS
- Internet connectivity for external API verification calls
- Browser: Chrome / Edge / Firefox for the Streamlit web interface

System Architecture

Layered pipeline from raw postings to explainable verdicts

1. Data Layer	Multilingual job postings from portals / APIs / datasets (EMSCAD, Kaggle)
2. Preprocessing Layer	Cleaning, tokenization, lemmatization, stop-word & language detection
3. Feature Extraction Layer	TF-IDF vectors, Bag-of-Words, contextual transformer embeddings
4. Machine Learning Layer	Random Forest / ANN / mBERT / XLM-RoBERTa classification + hierarchical clustering
5. Rule-Based & Verification Layer	Salary / email / keyword rules + government business-registration check (Verified / Not Verified) + external API checks (phone, domain)
6. Explainability & Evaluation Layer	SHAP / LIME explanations, accuracy / precision / recall / F1, confusion matrix
7. Recommendation & UI Layer	Real / Medium-Risk / Fake verdict, risk score, apply-or-avoid guidance via web app

Module Definition (I)

Data & analysis modules

1 Data Collection Module

Gathers job postings from public datasets (EMSCAD, Kaggle Balanced Fake Job Postings) and live recruitment platforms across multiple languages.

2 Preprocessing Module

Removes HTML, URLs, punctuation and duplicates; performs lowercasing, tokenization, stop-word removal and lemmatization; detects posting language.

3 Feature Extraction Module

Converts cleaned text into numerical form using TF-IDF / Bag-of-Words for classical models and contextual embeddings for transformer models.

4 Visualization & Clustering Module

Builds Tableau/Orange dashboards (pie, bar, choropleth, word cloud) and performs hierarchical clustering to reveal scam sub-types and geographic hotspots.

Module Definition (II)

Detection, verification & delivery modules

5 Classification Module

Random Forest / ANN / mBERT / XLM-RoBERTa models classify each posting as Real or Fake using extracted features; models are trained and evaluated with an 80/20 split.

6 Rule-Based, Government & External Verification Module

Flags free email domains, WhatsApp-only contact, unrealistic salary and urgency phrases; verifies whether the company is government-registered (MCA / GST / Companies House / SEC EDGAR) and cross-checks phone & domain via external APIs.

7 Explainability (XAI) Module

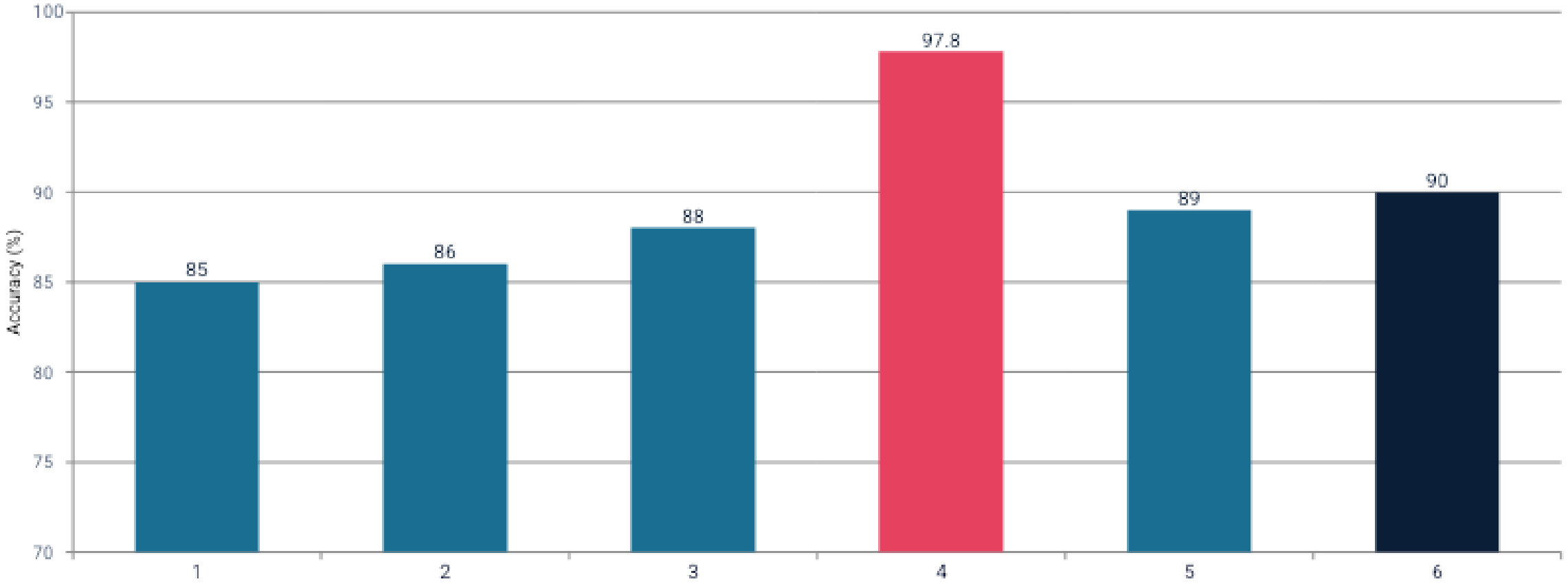
SHAP provides global feature importance; LIME explains individual predictions by highlighting the words responsible for the Real/Fake verdict.

8 Recommendation & UI Module

Displays final verdict (Real / Medium-Risk / Fake), fraud risk level, government verification status (Verified / Not Verified), suspicious features found, and apply/ avoid guidance through a Streamlit web interface.

Performance: Existing vs. Proposed

Accuracy trend across approaches reviewed in literature



Sources (upper-bound figures reported): Traditional ML & NLP-based / Deep Learning / Hybrid — JobShield comparison table; Transformer — Explainable AI multilingual study; Proposed Hybrid System — JobShield (87–90%).

Conclusion & Future Scope

- A unified hybrid pipeline — EDA/clustering + ML/DL + rule-based checks + explainability — outperforms any single approach in both accuracy and transparency.
- Explainable AI (SHAP/LIME) and external verification directly address the “black-box” and “no recruiter verification” gaps found in existing systems.
- Multilingual transformer support extends protection beyond English-only platforms to regional-language job seekers.

Future Scope

- Real-time job-portal integration & browser extension alerts
- Mobile app deployment for on-the-go verification
- Continuous learning from user feedback & emerging scam patterns
- Recruiter / company reputation scoring using verified databases
- Institutional deployment in university placement cells
- Integration with broader cybersecurity / phishing-detection tools

Thank You